

Determinant of Matrix:

It is defined as the sum of products of the elements of any row or column along with their corresponding co-factors. Determinant is defined only for square matrix.

It is denoted as Δ .

Cofactors of Element of Matrix:

Cofactors are related to minors by a small formula, for an element a_{ij} , the cofactor of this element is C_{ij} and the minor is M_{ij} then,

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

Ex Find the cofactor of the element placed in the first row and second column of the determinant:

$$\begin{bmatrix} 2 & 1 & 2 \\ 4 & 5 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

sol first row and second column element i.e. 1.

first find out the minor of 1.

$$\begin{vmatrix} 4 & 0 \\ 2 & 1 \end{vmatrix}$$

$$\therefore (4 \times 1) - (2 \times 0) = 4$$

$$M_{12} = 4$$

Now, cofactor formula.

$$C_{12} = (-1)^{1+2} M_{12}$$

$$C_{12} = (-1)^3 \times 4$$

$$C_{12} = -4$$

Determinant of Matrix 1×1

Let $X = [a]$ be the matrix of order one, then its determinant is given by

$$|X| = a.$$

Determinant of 2×2 matrix

The determinant of any 2×2 square matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}_{2 \times 2}$

is calculated using the formula.

$$|A| = ad - bc.$$

Determinant of 3×3 matrix.

It is determined by expressing it in terms of 2nd-order determinants.

It can be expanded either along rows (R_1, R_2 or R_3) or Column (C_1, C_2 or C_3).

Consider a 3×3 matrix A

$$A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix}$$

$$|A| = a_1 (b_2 c_3 - b_3 c_2)$$

$$- a_2 (b_1 c_3 - b_3 c_1)$$

$$+ a_3 (b_1 c_2 - b_2 c_1).$$